Faculdade de Design, Tecnologia e Comunicação
Universidade Europeia

Project 1 - CICLA
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Master in Creative Computing & Artificial Intelligence (IADE 2025–26)

MILESTONE 1

Research & Concept Development

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INTRODUCTION

This project responds to the CICLA challenge to design a connected, secure, and sustainable bicycle parking solution for Lisbon. Our concept, Eco Pod, builds upon this goal by integrating IoT sensors, AI prediction, and weather-aware feedback into a single interactive system.

While the main objective of this new product of Cicla is theft prevention, Eco Pod stands out by addressing the cyclist's emotional trust and comfort. We do this by anticipating needs like shelter from rain, real-time availability, and community feedback.

We hope to demonstrate how intelligent systems can enhance both urban mobility and user experience.

CICLA
Registo Nacional
de Bicicletas

ABOUT CICLA

CICLA is a Portuguese project that works to make bike ownership safer by registering and identifying bicycles, helping to reduce theft and create trust in urban cycling. With Eco Pod, CICLA expands into physical smart city solutions for public space, starting in Lisbon with 15 pods.

ABOUT PROJECT CONTEXT & SCOPE

This project is part of a **multi-disciplinary collaboration** between three Master's programs at IADE:

- **Creative Computing & Artificial Intelligence**, responsible for IoT tech, data visualisation, AI, and digital interaction.
- **Product, Space and Interaction Design**, physical structure, materials, space, environmental integration.
- **Design and Advertisement**, visual identity, communication & branding towards target audience and the local government as the stakeholder.

IADE
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ABOUT MILESTONE 1

This milestone (M1) focuses on **phase 1**, **research and concept development**. As the creative computing & AI team, we define the core idea behind the smart bicycle parking system for CICLA. This includes early research, the first concept proposal, & how AI, IoT, and user needs come together to shape the foundation for the prototype.

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THE "BIG PICTURE"

Urban Context

Lisbon is undergoing a major mobility transition. Under the MOVE Lisboa 2030 plan (CML, 2020), the city aims to build a **people-centred, low-emission mobility ecosystem**, expanding over **200 km of continuous cycling routes** and developing **secure bicycle-parking infrastructure** along the way. Between **2011 and 2017**, cycling trips in Lisbon increased by **over 200%**, yet cycling still represents *less than 1% of total travel* (MOVE Lisboa 2030, p. 39).

At a continental level, the **European bicycle market** is projected to grow at a *compound annual growth rate (CAGR) of 11.3% from 2025 to 2030* (Grand View Research, 2024).

While infrastructure expands, **everyday confidence has not caught up**.
A study published in 2019 found that cyclists in Lisbon faced major barriers including **perception of safety, physical effort, the lack of a safe cycling network, and bicycle ownership** (Félix, Moura, & Clifton, 2019).

This **persistent gap between policy goals and user trust** highlights the need for **connected, intelligent infrastructure** that makes cyclists feel *both protected and informed*.

Figure 1
Cycling network plan for Lisbon. Adapted from *MOVE Lisboa 2030: Sustainable urban mobility plan* (Câmara Municipal de Lisboa, 2020, p. 39).

CICLA's initiative

CICLA (CICLA, 2025) is a **non-profit organization** working to make cycling safer through **bicycle identification and registration systems**. It connects **riders, authorities, and communities** to prevent theft and improve accountability. The organization's next step is to **extend this security layer into public space**, creating **smart, connected parking pods** that align with Lisbon's "*smart city*" vision outlined in MOVE Lisboa 2030(CML, 2020).

Context & Problem Definition

WHAT'S MISSING

Problem Space Definition

Based on user research and client expectations, the following **challenges** have been detected:

- **Public parking** options *lack visible, trusted security*.
- **Weather exposure** *discourages long-term outdoor parking*.
- **Users** have *no real-time information about space availability*.
- The **system operator (CICLA)** needs *data to maintain and optimize pod performance*.
- **Existing infrastructure** can feel *static*. There is no interaction, no feedback.

Addressing these gaps requires not only design innovation but also *intelligent systems capable of sensing, predicting, and communicating in real time*.

Focus of Creative Computing & AI

For the Creative Computing and AI team, this challenge focuses on **how technology can build trust**, using IoT sensors, intelligent prediction, and interactive feedback to make parking safer, faster, and more transparent.
Our **goal** is to *design the digital and data-driven layer that supports the physical system* created by our partner teams in Product Design and Marketing.

"Security is the number one priority." — CICLA client meeting notes

Context & Problem Definition

WHAT'S NEEDED

These challenges translate into distinct needs and opportunities for each stakeholder involved in the EcoPod system.

Stakeholder	Needs	Opportunity
Cyclist	Safety, comfort, predictability	Real-time feedback, visible security
CICLA	Safety, Data for maintenance & trust	Sensor analytics, dashboards
City	Smart infrastructure, Sustainable mobility	Scalable, low-energy infrastructure

...the need for connected, intelligent infrastructure that makes cyclists feel both protected and informed.

User Research

UNDERSTANDING THE USERS

The EcoPod responds not only to Lisbon's infrastructural needs but also to the lived experiences of cyclists navigating the city every day.
The next section explores who these users are and what challenges shape their relationship with cycling in Lisbon.

User Focus

The Eco Pod concept primarily serves **Daily Urban Commuters**, cyclists who integrate biking into their work or study routines.
They need **speed, reliability, and visible safety** at public parking points close to offices, metro stations, or transport hubs.

A **secondary audience, Local Residents**, will benefit in later stages of deployment, particularly in neighbourhoods where private storage is limited.
For them, long-term trust and environmental protection (lighting, cover, durability) become key motivators.

First for Daily Urban Commuters

Then Local Residents

Both audiences share the same emotional driver: trust.
Without visible security and comfort, cyclists avoid public parking altogether,
a barrier EcoPod aims to remove.

User Research

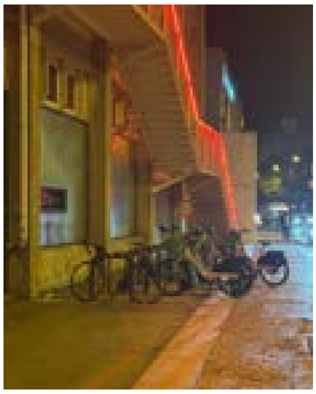
FIELD RESEARCH/OBSERVATIONS

Field visits across Lisbon (including Parque das Nações, Marvila, Cais do Sodré & Beato) revealed that cyclists rarely rely on public parking. Most prefer to bring their bikes indoors or keep them constantly supervised, reflecting a deep **lack of trust in street-level security**.
Existing racks are often shared with scooters, could be poorly lit, or exposed to weather, while covered parking is scarce and unevenly distributed across the city. The mix of steep terrain, narrow sidewalks, and irregular cycling lanes further discourages everyday riders from stopping or parking in public.
These conditions confirm that Lisbon's cycling infrastructure remains **fragmented and unpredictable**, reinforcing the emotional need for **safety, visibility, and reassurance** that shapes the EcoPod concept.

Marvila

Parque das Nações

Cais do Sodré



Beato

User Research

POSSIBLE INTEGRATIONS OF USER PAIN POINTS

These pain points, based on the briefing & research on daily urban commuters and field observations across Lisbon, connect real user frustrations to potential IoT and AI opportunities within EcoPod.

Pain Point	Design & Technology Implications for EcoPod
Fear of theft	Add *visible security cues* (RGB LED locks, confirmation lights) to reassure users.
Vandalism risk	Use *durable, tamper-resistant materials* and *sensor alerts* for unusual activity.
Weather exposure	Integrate *semi-covered structures* and *weather sensors* for protection and feedback.
Unpredictable space availability	Apply *AI prediction* to show real-time space status in an app interface.
Poor lighting and visibility at night	Add *guided LED lighting* and *clear visual cues* for easy access.
Limited digital feedback after parking	Connect *IoT sensors* to a user app and operator dashboard for live status and maintenance data.

For the **next phase**, only **key features** will be developed to ensure a **focused and feasible prototype**.

User Research

CROSS-MASTER REFERENCE

These findings complement the user research conducted by the *Design & Advertising* team (Gonçalves, 2025), ensuring a shared understanding of Lisbon's cycling challenges across master programmes.

USER RESEARCH CONCLUSION

For the *Creative Computing & AI* team, this research highlights that **trust** is the emotional core of cycling behaviour, a principle that will guide every technological layer of EcoPod, from sensor feedback to predictive monitoring.

In other words, the research clarifies that technology must first and foremost **build trust**. Every data interaction, light signal, or app feature should reassure the cyclist that their bike is **safe, accessible, and cared for by the system**.

RESEARCH

Market Research

Shared from the Design & Advertisment team

COMPETITOR ANALYSIS & BENCHMARKING

Building on the user insights, this section examines how current market solutions tackle similar challenges. Based on research by the Design & Advertising team (Gonçalves, 2025), it highlights where EcoPod can bring innovation.

Competitor	Strengths	Weaknesses	Lessons for Eco Pod
Bikeep	High-tech lockers, smart access	Expensive, rigid	Keep app simple and onboarding fast
Oonee Pods	Modular, community-driven design	Large footprint	Prioritize compact design
Cyclehoop *(UK)*	Tested urban models	Minimal digital layer	Combine physical reliability with digital UX
EMEL BiciPark *(Lisbon)*	Municipal trust, accessible	Limited tech, low perceived safety	Pair municipal legitimacy with real-time feedback

OPPORTUNITY AREA

We can expand *beyond anti-theft design* to *integrate real-time AI predictions, weather-aware insights, and user feedback loops*.
By merging IoT data with user-centered experience design, it transforms parking *from* a *static* structure *into an interactive experience* that helps encourages people to ride their bikes.

Existing solutions secure bikes. Eco Pod secures cyclists' trust.

Key Insights & Opportunities

SYNTHESIS

Although cycling in Lisbon is steadily growing, driven by sustainability policies and the rise of e-bikes, self-owned cyclists still face persistent barriers.
Research conducted by the *Design & Advertising* and *Creative Computing & AI* teams identifies theft concerns, weather exposure, and low trust in public parking as the main obstacles to daily use. These findings define the technological priorities for EcoPod and guide our focus for concept development.

Security Anxiety

Cyclists avoid street parking entirely. Most bring bikes indoors or keep them constantly supervised.

Weather Exposure

Rain and sun damage discourage long-term outdoor use of uncovered parkings.

Unreliable Infrastructure

Inconsistent lighting, signage, and access make current parking options confusing or unsafe.

Predictability & Visibility

Commuters want to know in advance if a pod has free space, and they need visible cues that confirm their bike is safely locked.

Not all insights are fully addressable through technology.
See next page for feasibility focus and team priorities.

Technology follows insights.
Cyclists don't just need parking.
They need to feel safe and confident using it.

Key Insights & Opportunities

FEASIBILITY FOCUS & TEAM PRIORITIES

By mapping each insight against our Creative Computing & AI scope, we identified which ones are realistically addressable for our concept development.

Insight	Feasibility	Possible Tech Approach / Focus
Security anxiety	High	IoT sensors, motion detection, light feedback
Predictability & visibility	High	Occupancy sensing, AI prediction, real time display
Unreliable infrastructure	Partial	Pod visibility, interface guidance, data logging
Weather exposure	Limited	Context-aware weather updates for the comfort of the user

RESEARCH CONCLUSION

While environmental and infrastructural issues extend beyond our scope, technology can still make *safety* and *trust* visible through data, sensing, and feedback.
These findings bridge cyclists' needs with CICLA's and the city's goals, setting the stage for concept development in the next phase.

By defining these priorities early, EcoPod can evolve from research into an implementable concept. One that starts small but builds real trust.

Key Insights & Opportunities

FROM INSIGHTS TO OPPORTUNITIES

Eco Pod is designed primarily for **daily urban commuters**, later extending to local residents who rely on street-level parking.
While centred on cyclists' experience, it also supports **CICLA's mission** to promote safer cycling and the **City of Lisbon**'s goal of building a scalable, low-energy mobility network.
Building on the briefing and research findings, the following goals outline what users and stakeholders expect from a smart parking system and how technology can make it real.

Primary User Goals

1. Park and secure a bike quickly with minimal steps.
2. Check space availability before arriving.
3. Avoid leaving the bike in exposed or unsafe areas during bad weather.
4. Receive reassurance that the bike is safely locked.
5. Contribute quick feedback to improve shared safety and trust levels.

Primary Stakeholder Goals

CICLA - Client Perspective

User App
1. Connect to CICLA's national bike registry dataset or other for user login to simulate pre-requirement of bicycle registration before use of app.
2. Let users check real-time pod availability, book a space, and unlock it.
3. Provide clear digital feedback confirming lock status and occupancy.
4. Encourage CICLA's core value of community in app.

Back-Office Dashboard
1. Monitor real-time pod occupancy and user flow.
2. Track parking duration to prevent long-term misuse and automate fine warnings.
3. Integrate data to maintain the hive of pods (detect malfunctions, plan repair, extend a pod,...).
4. Keep the system modular, low-cost, and easy to expand city-wide.

City of Lisbon - Urban Perspective

1. Align with Lisbon's MOVE 2030 vision by developing smart, low-energy systems that support daily urban cycling.

Key Insights & Opportunities

Design Opportunities

These opportunities mark the bridge between research and concept development. Each one connects a user and/or stakeholder need to a creative computing response showing a start on how IoT, AI, and interactive feedback can support cyclists, assist CICLA in monitoring performance, and align with Lisbon's smart-city & urban mobility goals.

USER / STAKEHOLDER NEEDS	PAIN POINTS	OUR SOLUTION
Security & Trust	Fear of theft or vandalism	Visual lock confirmation (RGB LED), secure data link with CICLA bike registry
Predictability	Uncertainty about available space	AI-powered occupancy prediction in the app
Comfort	Weather exposure discourages use	Weather-aware guidance + covered pod design
Speed & Access	Time lost searching or unlocking	One-step RFID/NFC check-in and automatic lock detection
Visibility & Clarity	Confusing layouts or poor lighting	Interface and RGB LED cues for guidance, clear digital feedback
Feedback & Community	Lack of connection or reassurance	User rating system that updates comfort scores dynamically

CONCEPT DEVELOPMENT

The Idea

CORE CONCEPT

Eco Pod was born from a simple observation: cyclists in Lisbon don't just need a place to park, they need a reason to trust it. Designed for daily commuters first, EcoPod transforms ordinary street parking into a connected experience where technology builds confidence in safety and reliability.

Using IoT sensors, live feedback, and contextual data, the system reassures users that their bike is safely locked and that space is available before they arrive.
In doing so, Eco Pod becomes more than infrastructure.
It acts as a **digital layer of trust in everyday parking**, a small yet meaningful step toward safer, smarter urban cycling.

RESEARCH TO CONCEPT FOCUS

Our research made one truth clear: the barriers keeping people from cycling daily aren't just physical, they're emotional. Fear of theft and uncertainty about parking availability discourage everyday use.

From the four key insights *(security anxiety, weather exposure, unreliable infrastructure, and predictability & visibility),* our team defined which could be realistically addressed through technology.
For Milestone 1, Eco Pod focuses on **security reassurance** and **predictability**, supported by early integrations of **weather awareness** and **occupancy sensing**.

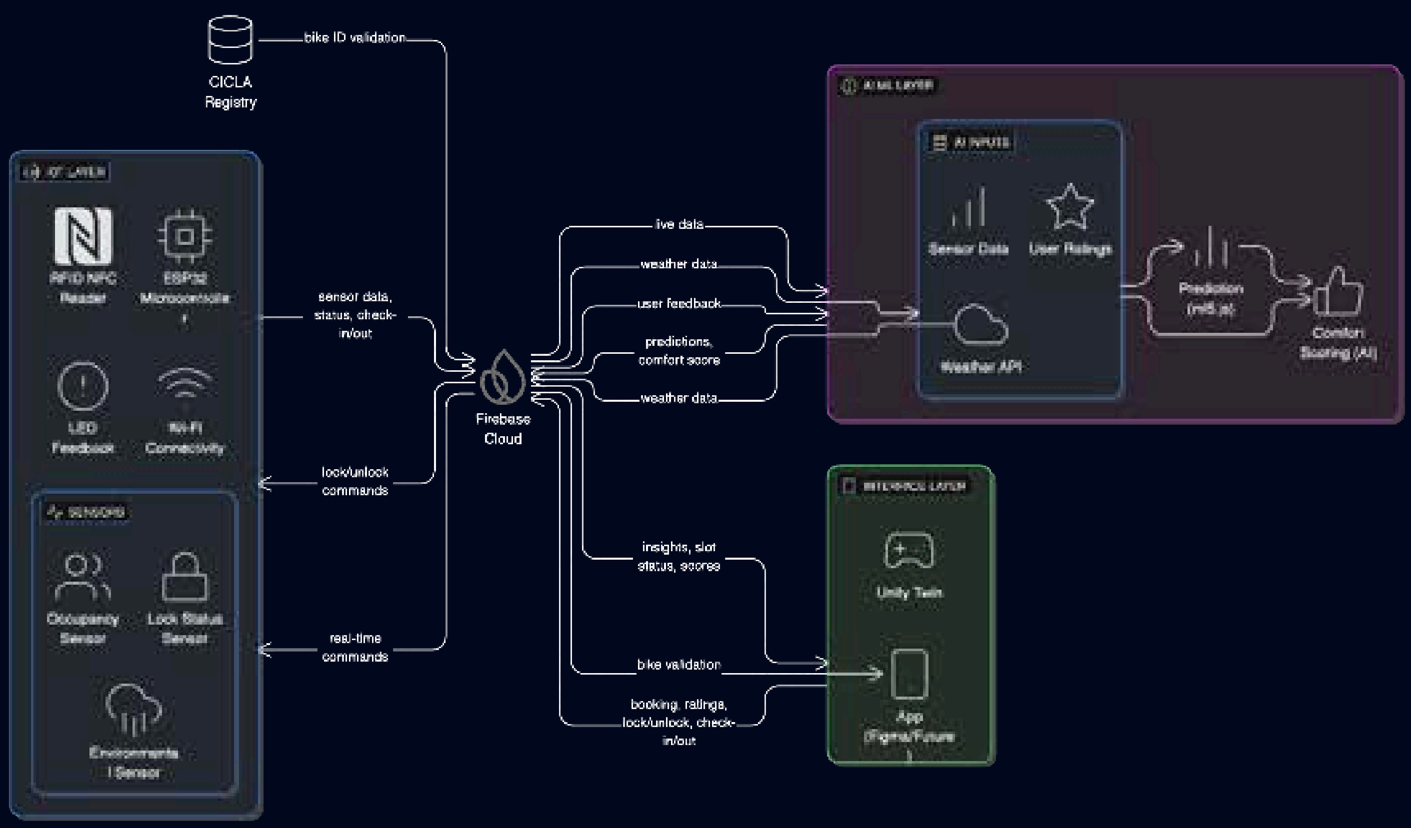
This means exploring how **IoT sensors and light feedback** can make locking actions feel trustworthy, how **AI predictions** can reveal parking availability before arrival, and how **simple weather data** can guide cyclists toward safer, covered options.
Instead of building full mechanical systems, we start with the **digital foundation**, turning reassurance into visible, measurable interaction.

This focused and testable scope connects human trust with technical feasibility and sets the foundation for the next phase: **Prototyping and Technical Exploration (Weeks 4 – 8)**.

System Overview

Eco Pod is structured across three complementary layers:

IoT, AI/ML, and Interface



Instead of implementing full barometric and wind sensing, Eco Pod integrates weather APIs.
This reduces hardware cost and complexity while still allowing us to use weather conditions as an input to our AI models.
Additional sensors, such as light or temperature, can be integrated later as the physical design and prototyping evolve and budget allows.

System Overview

WHAT ABOUT DATA?

The initial data flow establishes how EcoPod connects physical sensing, environmental context, and user reassurance into one smart system. Occupancy, lock status, and weather data form the foundation for intelligent interaction, detecting activity, predicting availability, and adapting to conditions.
A fourth data flow in the future, user feedback, will complete the loop by adding community trust data to refine comfort scoring and improve reliability.

Data Flow 1: occupancy level of pods

Shows live pod usage and feeds the Machine Learning predictor with pod history (historical occupancy data, time of day, real time weather conditions affecting pod).
(#Physical Computing, ML, Creative Programming)

IoT sensor (load cell/ IR) → ESP32→ Firebase → App + ML model

Data Flow 2: lock status of parking slots

Provides visible reassurance to users (LED + app icon).
(#Physical Computing, AI, Creative Programming)

RFID/NFC input → Lock sensor (reed/limit switch) → ESP32 → Firebase
↓
App UI
& AI logic
+
LED RGB output

System Overview

Data Flow 3: weather data

Adds environmental awareness to predictability & comfort.
(# Introduction to AI, Applied ML)

Weather API (or simulated data) → AI Comfort Score / ML Feature

Data Flow 4 : user data (optional)

Integrates community trust and emotional safety into system intelligence; not MVP, initial core concept but planned for later.
(#AI, Creative Programming)

App user input / pod rating system → Firebase → AI Comfort Score

User data history → Firebase → AI with ML Comfort Score

AI + ML Integration

THE BROADER IDEA

The AI/ML components are designed to:

- **Predict pod availability** based on historical occupancy data, time of day, and weather conditions.
- **Compute "Comfort Scores"** that reflect both physical (rain, temperature, coverage) and social factors (user ratings).
- **Personalize recommendations** by suggesting covered pods or safer areas during poor weather conditions.

This creates a learning loop between the user, the environment, and system:

Data → Prediction → Feedback → Refinement.

The predictive component reduces user uncertainty ("Will I find space?"), while the comfort-scoring logic adds emotional reassurance ("Is my bike safe and dry?"). Together, these features shift Eco Pod from a functional tool to an intelligent urban companion for cyclists.

AI vs ML Roles

Component	Description	Role
AI logic (Comfort Scoring)	Combines user feedback, weather, and history	Improves perceived safety. "If rain > 60% → recommend covered pod."
ML Model (Occupancy Prediction)	Learns parking patterns over time using ml5.js	Helps forecast available pods
IoT System	Captures real-time slot and lock data	Feeds ML and app updates

AI = systems that act intelligently using logic, scoring, or decision rules.
ML = systems that learn patterns from data.

CONCEPT DEVELOPMENT

Artificial Intelligence

AI LOGIC: MAKING SAFETY AND PREDICTABILITY VISIBLE

GOAL
Visualize and simulate how an intelligent pod mapping system could guide cyclists toward safer, available parking. Improves perceived parking safety of the user ("Is my bike safe and dry?").

CONCEPT

EcoPod uses AI logic in p5.js to simulate intelligent urban pod mapping behavior.
The client mentioned that for real-world testing, the idea is to launch a hive of 15 Eco Pods across Lisbon that are nearby each other and form a connected network.
The design and advertisement team point out a gradient line map.
This gave us the idea of mapping the pods for the user with AI layer over it that implement comfort scoring.
For an initial concept, this means **reactive and generative AI agents** in **P5.js** responding to live or simulated data. The agents respond to **weather and real time pod occupancy status conditions,** (and optional user data) creating a live *user comfort map* of the pods in Lisbon, that visualizes perceived safety and environmental comfort. For Milestone 1, this represents the first conceptual bridge between environmental data and user trust.

Moodboard by Design & Advertising team (Gonçalves, 2025)

AI LOGIC: MAKING SAFETY AND PREDICTABILITY VISIBLE

Agent Types

<table>
<tr><th>Agent Type</th><th>Function</th><th>Data Required</th></tr>
<tr><td>Generative Agents</td><td>Visually shape the “comfort map field”, showing safer or more pleasant pod zones to park.</td><td rowspan="2">1. Add weather api data
2. Add pod occupancy data
3. Add user feedback (optional)</td></tr>
<tr><td>Reactive Agents</td><td>“Cyclists” move toward or away from pods depending on real-time comfort conditions (e.g., rain, space availability) to give user a real time mapping of the pods.</td></tr>
</table>

These are overall data requirements. Defining the specific used data types will be more known in the next phases of the project since prototyping and code experiments are still in shaping the outcome.

Applied Machine Learning

ML MODELLING: LEARNING FROM DATA TO ANTICIPATE USE

GOAL

- Predict pod availability and reduce uncertainty ("Will I find a space when I arrive?"),
- Help CICLA understand the pod usage trends for future planning.

CONCEPT

A lightweight **ML model** is proposed to forecast how likely each pod is to be available. It's trained in **ml5.js** (ml5 NeuralNetwork) using a dataset that combines simulated IoT and weather data.

DATA PIPELINE

IoT Sensors & Weather Api → Firebase Database → ML Model (ml5.js) → App/Dashboard

DATA SOURCES

Feature	Description
Pod ID	Which pod is it?
Weather API Data (rain, sun, heatwave, temperature)	Affects likelihood of cycling
Occupancy History (from Firebase / simulation)	Core predicting indicator of future demand
Time of Day / Weekday	Real time data

These are overall data requirements. Defining the specific used data types will be more known in the next phases of the project since prototyping and code experiments are still in shaping the outcome.

Applied Machine Learning

LEARNING FROM DATA TO ANTICIPATE USE

Expected Outputs

Output	Purpose
Occupancy Probability (0–1 or 0-100%)	Predicts how likely the pod has a free space
Confidence Score (%)	Indicates reliability of the prediction
Data Visualization (map / dashboard)	Displays availability/occupancy forecast to user or operator(=client)
Cycling Parking Occupancy Classifications (optional)	To make cycling parking behaviour patterns out usage & occupancy of the pods (mainly for the client) (e.g. commuting or residential parking?, short or long term occupancies, long-term misusers that need to be fined?)

These output will be tested out in the next phases of the project. Iterations can still take place depending on the data/dataset use, concept alterations and alignment with primary goals set in research.

Implementation Notes

- For Phase 2, the model will first be trained on a small simulated dataset.
- In later phases, ML logic will be connected to the user app (space availability) and back-office dashboard (usage trends, maintenance insights).
- Real weather sensors are replaced with a weather API to reduce cost and simplify maintenance. Testing in Phase 2 will reveal if this is a good alternative.

Creative Programming

FRONT OFFICE: USER APP

GOAL

- Help cyclists feel informed and reassured at every step, from finding a space to confirming that their bike is safely locked.
- Meet Primary User and Client Goals
- Shape the User Experience on the app with Eco Pod concept

CONCEPT

A p5.js prototype simulates the EcoPod mobile app, integrating the core concept challenges:

- perceived safety (visual lock feedback)
- AI comfort scoring and space predictability
- community trust input (basic feedback placeholder)

The focus is on mapping the experience, not building a full app — showing how data from IoT sensors, weather APIs, and AI predictions can be represented in an interactive user flow.

CONCEPT DEVELOPMENT

Creative Programming

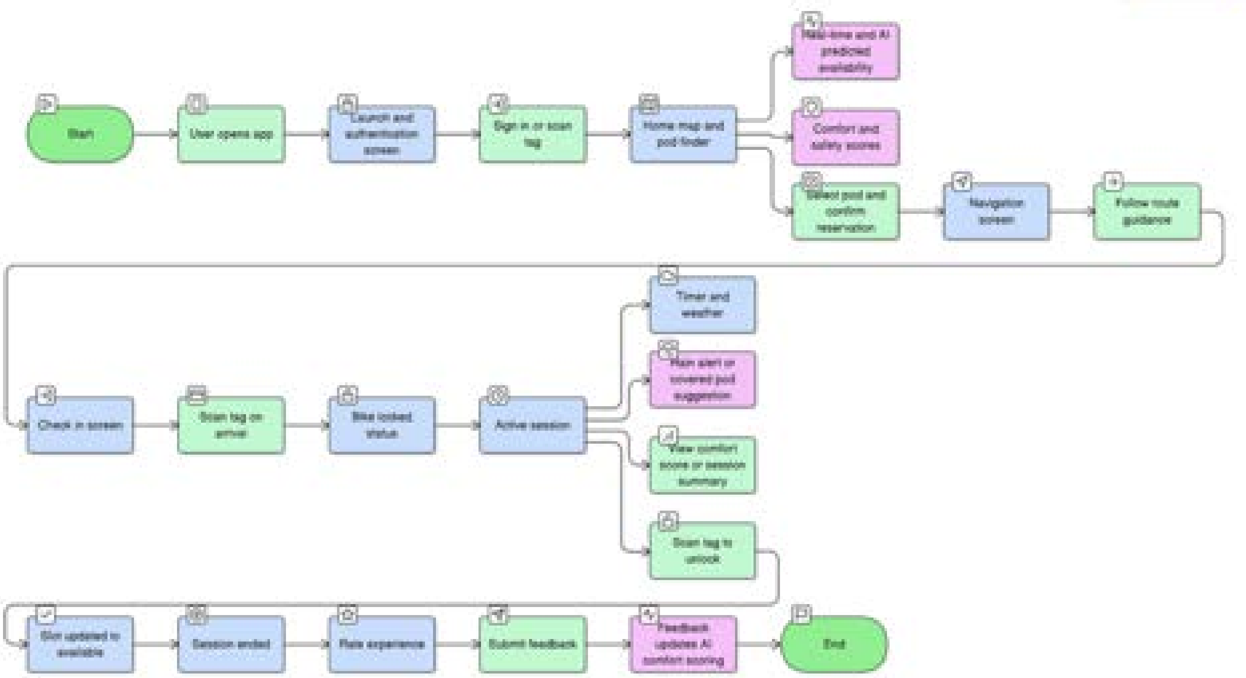
USER EXPERIENCE

After identifying key user pain points around trust, safety, and unpredictable weather and primary goals, the app flow demonstrates a typical EcoPod journey:

1. Login / Authentication (simulated connection to CICLA registry)
2. Check availability (AI occupancy prediction + live data)
3. Book or unlock pod
4. Receive lock confirmation & comfort feedback (color cues, weather awareness)
5. Submit quick feedback / trust rating

For Milestone 1, this will be represented through static screens or simple p5.js interactions rather than a functional system.

User Flow (App Interaction Concept)



This diagram illustrates the main in-app journey for cyclists using Eco Pod, from logging in to completing their parking session. It shows how weather-aware AI predictions and user feedback integrate into the interface. An initial booking system flow has been integrated.

BACK OFFICE: CLIENT DASHBOARD

GOAL

- Give CICLA visibility over pod usage, occupancy, and maintenance, turning data into operational insight.
- (Meet primary client goals)

CONCEPT

The back-office system allows CICLA to monitor Eco Pods activity, visualizing data (occupancy, lock status) to manage pods more efficiently.

For Milestone 1, this concept of the client dashboard was focused first on the making of the digital twin of the Eco Pod product itself. Later data visualizations and dashboard interface will be added.

Creative Programming

UNITY SIMULATION & CLIENT VISUALIZATION

To demonstrate Eco Pod from a client perspective, we will build a Unity-based digital twin of the parking system. The Unity scene represents a small Lisbon street segment with several Eco Pods, each containing two parking slots.

Each pod is modeled as an interactive object with three main visual states: Free, Occupied, and Locked.

Real-world variables from the IoT prototype (slot occupancy, lock state, RFID check-in/out) are mapped to simple visual cues in Unity:
bikes appear or disappear in slots, RGB LEDs change color, and lock icons toggle between open and closed.

A side panel dashboard shows per-pod Comfort Scores, predicted availability labels, and a compact weather summary.

During the presentation, this simulation allows CICLA and other stakeholders to see how the system behaves over time without needing to inspect the electronics directly.

A short scenario will illustrate a full cycle:

1. a cyclist reserves a pod,
2. checks in on arrival,
3. locks the bike (visualized by the pod changing state),
4. receives rain-aware guidance, and later checks out and rates the experience.

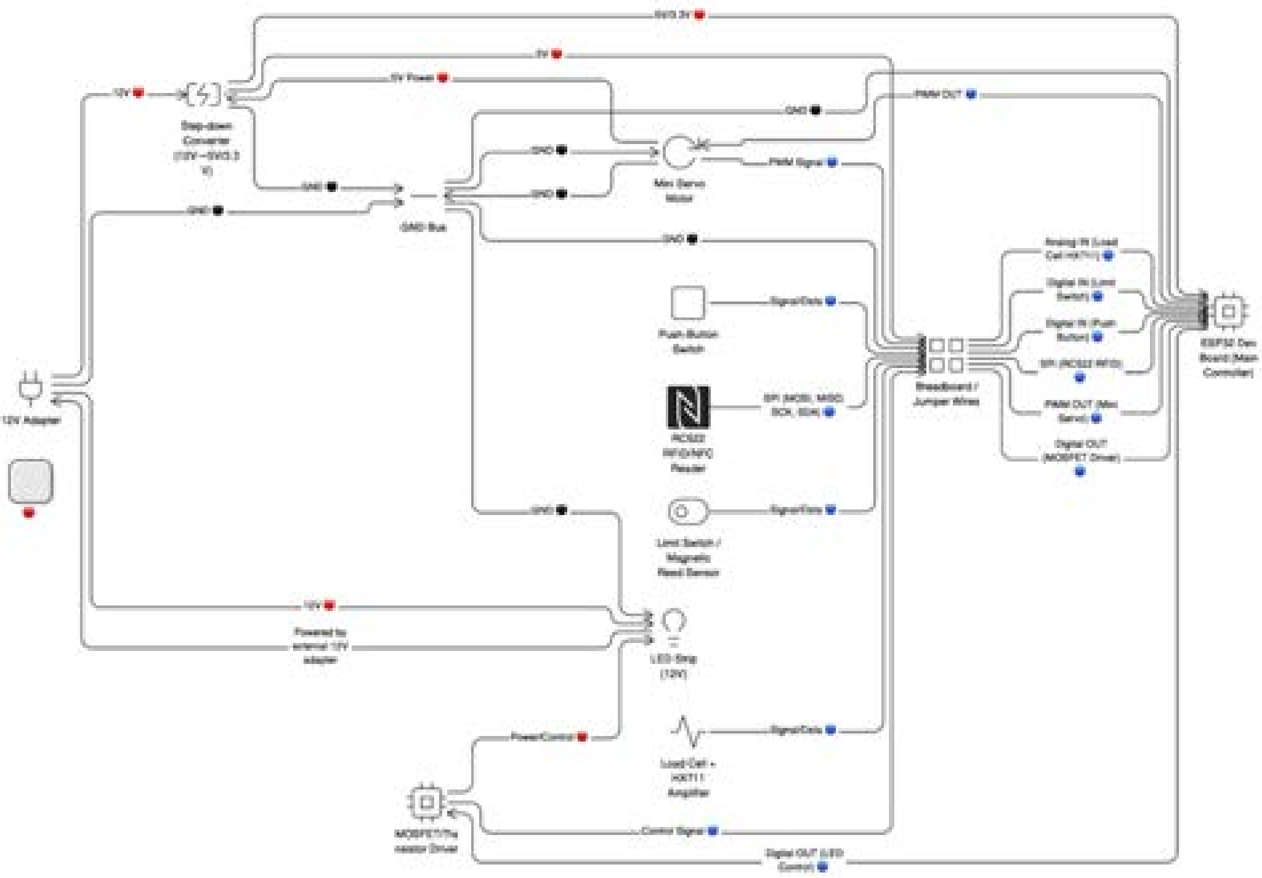
In early stages, these events will be triggered through scripted sequences; in later iterations, the Unity scene will listen to real-time updates from Firebase so that the digital twin reflects the actual hardware prototype.

This approach turns Eco Pod into a two-layer prototype: the physical IoT build that proves technical feasibility, and a Unity "client view" that communicates system value and behavior in a clear, engaging way.

Physical Computing & IoT

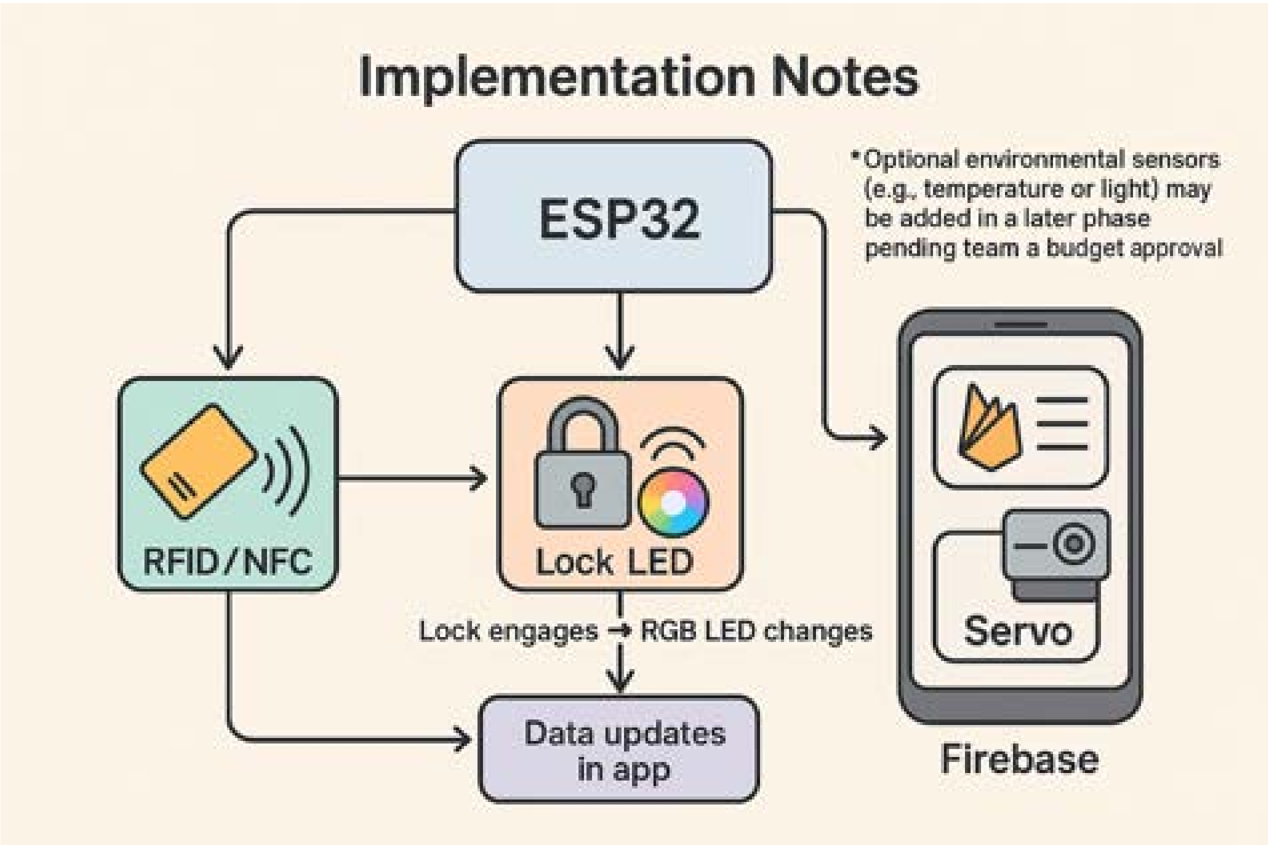
HARDWARE SCOPE

For the Eco Pod prototype we focus on a small number of highly relevant sensors: occupancy (load cell or IR), lock status (limit switch), controlled by an ESP32 microcontroller and visualized through RGB LED feedback. RFID/NFC tags and a reader provide a bike-centric identity layer connected to the CICLA registry.
This configuration is enough to support our main IoT interactions (park, lock, identify the bike) without reproducing a full industrial weather station. Weather data will instead be obtained via an external API and used in our AI and ML logic.



A list of material and budget proposal for initial physical prototyping can be found in the Appendix.

Physical Computing & IoT



Implementation Notes

- The ESP32 will control sensors, RGB LEDs, and servo, sending data to the Firebase database for visualization and app integration.
- RFID / NFC check-in forms the main interaction demo: tag scan → lock engages → RGB LED changes → data updates in app.
- Optional environmental sensors (e.g., temperature or light) may be added in a later phase pending team and budget approval.

Next Steps

SYSTEM INTEGRATION & DATA FLOWS

- Add Data Flow 4: user data
- Ensure Data Flows for stakeholder's dashboard and back office

AI & ML

Future AI
- Integrate User data in "comfort mapping"
- Extend Comfort Mapping idea with Model Based Agent

Agent Type	Function	Data Required
Model-based Agent (Next Step)	Retain memory of previously safe or comfortable pods, improving user route decisions over time.	Historical pod performance
Model-Based Extension (Next Step)	Learn from previous "safe" zones.	Historical comfort data

Future ML
- If useful, cycling parking occupancy pattern classifications can be tested out and made for the client.
- In later phase: integrate ML models in user app and client dashboard.

Examples of brainstormed AI/ML Ideas that didn't make it
- Identify unusual activity or repeated failed lock events (possible theft).
- Recommend maintenance based on abnormal sensor data.
- A bike wheel pattern recognition for quick access identifying as a user to park your bike.

CONCEPT DEVELOPMENT

Next Steps

CREATIVE PROGRAMMING

Front office: User App

- Refine the interface with clear feedback and navigation logic.
- Explore how community scoring affects the comfort index.
- First test integration with simulated AI map of pods integration for workable prototype.
- If time, add working AI-based predictions and weather updates.
- If time, implement real-time data exchange with Firebase and IoT prototype.

Back office: Client Dashboard

- Design Dashboard
- Prototype Dashboard Experiments

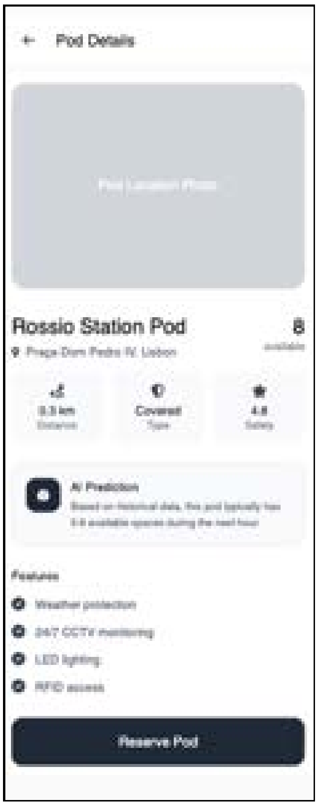
PHYSICAL COMPUTING & IOT

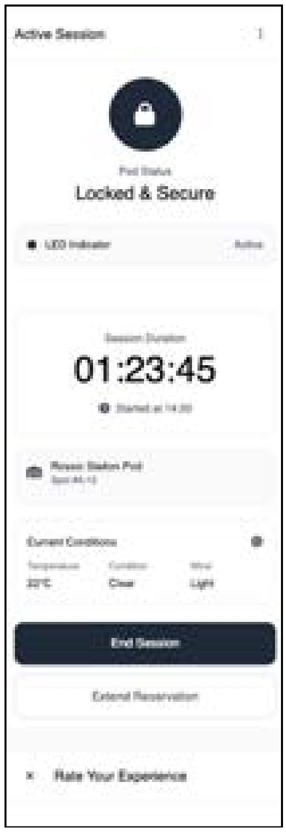
- Workable prototyping
- Extend physical computing and IoT: Add lighting output for navigation of pod in the dark to strenghten physical product and safety
- In later phases: ensure a concept with a network of pods to be operational

PROTOTYPES

App

APP INTERFACE WIREFRAMES



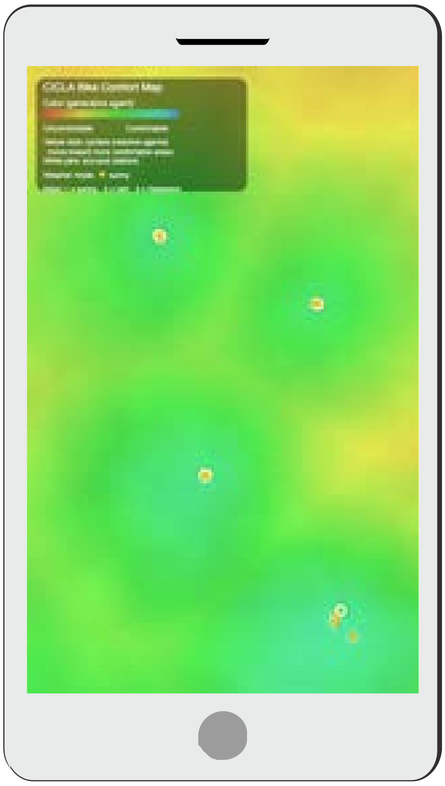


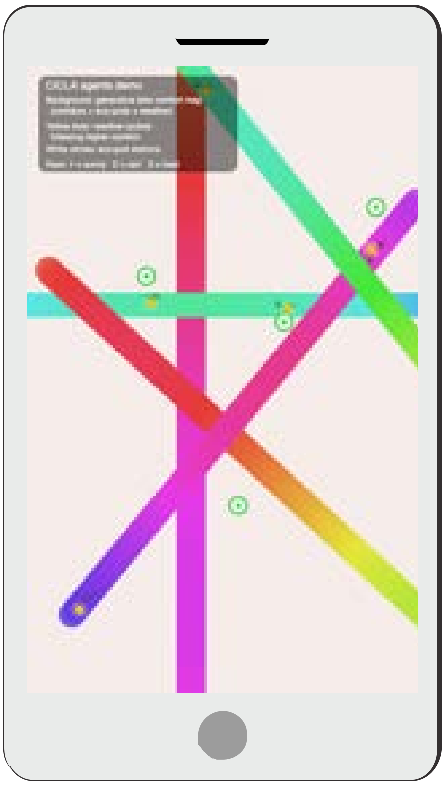
PROTOTYPES

App

P5.JS SKETCH & AI AGENT TESTS

We wanted to test how we could apply AI to the creative visuals. Here are two interactive prototypes that explore how generative and reactive AI agents could visualize real-time cycling comfort in Lisbon.
The generative agent creates a dynamic "comfort map" influenced by environmental factors such as weather, temperature, and proximity to eco-pods. The reactive agents simulate cyclists responding to those changing conditions, moving toward safer and more comfortable routes. Further experiments are needed to design this for either the user, or the backend for the client.

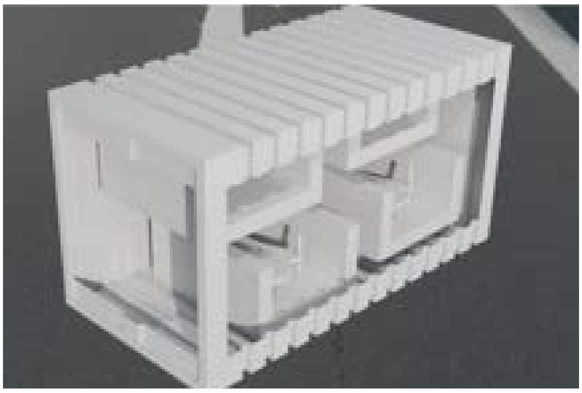




Next steps include integrating live data (weather APIs and sensor inputs) and refining the visual language to communicate comfort levels and flow patterns more clearly for both users and city planners.

PROTOTYPES

Product Concepts



Prototype 1 :
“The Box”

- Designed and modeled in Blender by Andrey.
- Concept based on recycling of used ship containers. Suitable for storage Scooters and bikes, while protecting them from both weather and vandalism.
- Concept approved by the Creative Computing x AI team.
- Declined by Product Design due to high production cost and complexity. Too many mechanical parts to fabricate.

Prototype 2:
“Vertical Capsule”

- Iterated into a more lightweight, tree-like vertical structure.
- Reduced materials, improved user access, and introduced solar panels for sustainability.
- Inspired by the Fibonacci sequence, symbolizing organic growth and harmony with nature.
- Improved accessibility because of automated clamp-hanging system.

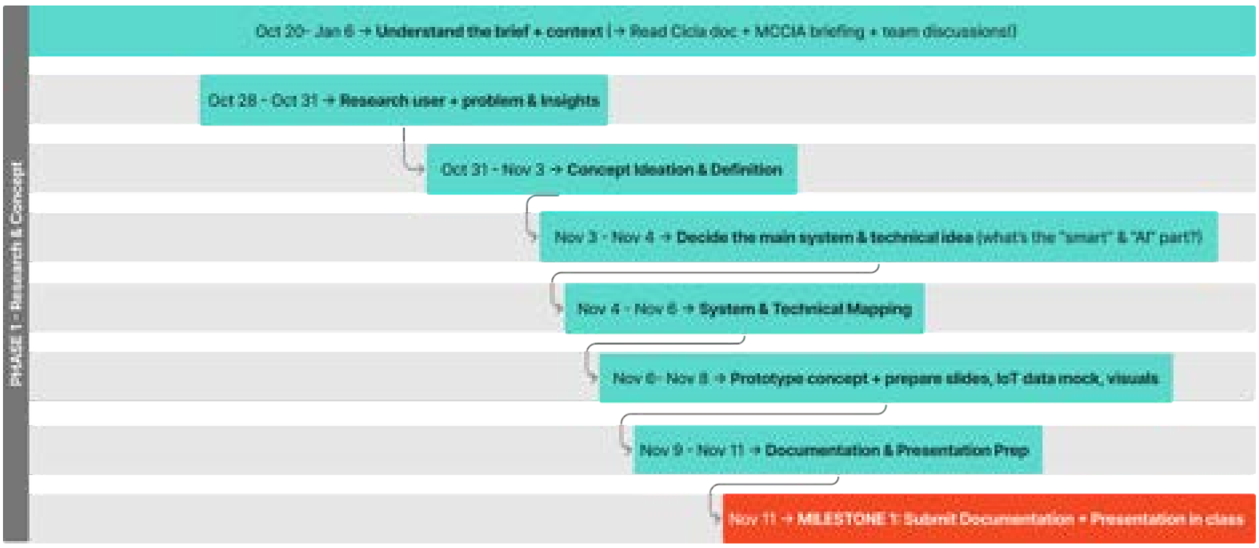
Next Steps:
“Prototype 3” (in progress)

- Following client feedback on 4 November: vertical storage may not be ideal for daily users.
- Our team will focus on a “technology first” solution that is not dependent on the physical end model.

Execution Record

ROADMAP AND TASKS

The initial plan for this milestone 1 was captured in a Gantt timeline.



Due to client meeting on Nov 4 and internal group changes on nov 5, going from three team members to four team members the execution on this plan has changed.

At first a Figjam File was used to cover task management with lists, but since that was not completely clear for every group member, this changed. Towards the end of milestone 1 and in the preparation of milestone 2, a github project is started up to divide better tasks among each other and per milestone.

TEAM

Workload Distribution

MILESTONE 1

For milestone 1 the workload was first assigned among 3 team members first.

Team Member	Role / Focus Area	Main Responsibilities	Deliverables / Outputs	Progress Evidence
Andrey Dyakov	IoT & Unity Prototype Lead	• Develops hardware concept and sensor integration using ESP32. • Defines physical data points (occupancy, power, charging). • Builds early Unity model of the capsule to visualize interactions. • Coordinates with Product Design on spatial dimensions.	• Sensor diagram and data table. • Photos/videos of ESP32 tests. • Unity prototype (opening capsule).	• Prototype photo/video. • Unity scene file. • Notes on hardware logic.
Nadine Allan	Creative Computing & Documentation Lead	• Writes all technical documentation, report sections, and slides. • Designs p5.js visualization interface for data and AI prediction mockup. • Defines AI/ML logic and data flow diagram. • Organizes FigJam board and milestone tracking.	• AI concept write-up. • p5.js visualization demo. • Integration diagram. • Final presentation slides.	• GitHub/p5.js file. • Report draft. • Organized FigJam documentation.
Sophia	Research & Partner Context Support	• Researches existing smart parking solutions and sustainability trends. • Summarizes Cicla's goals and key pain points. • Helps describe context and relevance for report sections. • Reviews and edits report for clarity. • Drafts "Problem & Why It Matters" section for dossier	• Research summary (1–2 pages). • Partner analysis notes. • Sources and references list.	• Google Doc notes. • Reference citations. • Text contributions in final report.

Due to extra team member, the workload was redistributed to adapt and to use what both teams had from nov 5th on.

Andrey => IoT & Unity Prototype
Nadine => Creative Computing, Interaction design, Presentation, Info collector
Inge => Research & Documentation

Workload Distribution

MILESTONE 1

The final workload turned out like this.

Team Member	Role / Focus Area	Main Responsibilities	Deliverables / Outputs	Progress Evidence
Andrey	IoT & Unity Prototype Lead	• Develops hardware concept and sensor integration using ESP32. • Defines physical data points (occupancy, power, charging). • 3D Modelling of Product. • Coordinates with Product Design	• Circuit diagrams and data flow. • Unity product prototypes photo/ video • Finalize physical computing and IoT documentation • List of Requirements	• Prototype photo/video. • Unity scene file. • Notes on hardware logic.
Nadine	Creative Computing, Interaction design, Presentation	• Writes all technical documentation, report sections, and slides. • Design AI p5.js visualization interface for data and AI prediction mockup. • Organizes team • Coordinates with marketing/brand team to develop visuals for app • Figma: add concept and system, prototyping and build drafts	• System Integration Flow • Start AI /ML concept write-up for documentation. • p5.js visualization demo. • Integration diagram. • Final presentation slides.	• GitHub/p5.js file. • Report draft. • Organized FigJam documentation.
Inge	Research & Documentation	• User Research • Documentation • Figma organization • Defines AI/ML logic and data. • Organizes team and tasks	• Do Research and brainstorm (see figjam) • Finalize Research Documentation, AI & ML, Team Chapter and References • Organize figjam file	• Github repo • Research & Brainstorm in Figjam • Documentation finalisations

TEAM

Workload Distribution

FURTHER PROJECT VISION

In addition to the Milestone 1 workload distribution, which covered the specific tasks for this phase, the following table defines the ongoing roles each member will maintain throughout the entire project. These roles clarify who is responsible for each course area and who will answer related questions in presentations or for reviews.

Team Member	Role / Focus Area	Main Responsibilities	Deliverables / Outputs	Progress Evidence
Andrey	IoT & Unity Prototype Lead	• Develops hardware concept and sensor integration using ESP32. • Defines physical data points (occupancy, power, charging). • 3D Modelling of Product. • Coordinates with Product Design	• Circuit diagrams and data flow. • Unity product prototypes photo/ video • Finalize physical computing and IoT documentation • List of Requirements	• Prototype photo/video. • Unity scene file. • Notes on hardware logic.
Nadine	Creative Computing, Interaction design, Presentation	• Writes all technical documentation, report sections, and slides. • Design AI p5.js visualization interface for data and AI prediction mockup. • Organizes team • Coordinates with marketing/brand team to develop visuals for app • Figma: add concept and system, prototyping and build drafts	• System Integration Flow • Start AI /ML concept write-up for documentation. • p5.js visualization demo. • Integration diagram. • Final presentation slides.	• GitHub/p5.js file. • Report draft. • Organized FigJam documentation.
Inge	Research & Documentation	• User Research • Documentation • Figma organization • Defines AI/ML logic and data. • Organizes team and tasks	• Do Research and brainstorm (see figjam) • Finalize Research Documentation, AI & ML, Team Chapter and References • Organize figjam file	• Github repo • Research & Brainstorm in Figjam • Documentation finalisations

TEAM

Teamwork & Synergy

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APPENDIX

LIST OF FIGURES

LIST OF REQUIREMENTS

Functional Requirements

Use Case	Step-by-Step Flow (HOW the user interacts)
1. Register & Log In	1. Open app → "Continue with Google" or "Apple ID" (OAuth) 2. App receives ID token → verifies with backend via HTTPS 3. First-time user: Consent screen ("Allow location for locker discovery?") → Save preferences locally 4. Home screen loads with map of nearby lockers
2. Find & Reserve a Locker	1. Map auto-centers via coarse location (Wi-Fi + cell) 2. Tap locker icon → shows real-time slots (via BLE beacon broadcast) 3. Tap "Reserve 30 min" → app sends encrypted reservation token to locker via BLE 4. Locker LED blinks green + push: "Slot 3 reserved!"
3. Load Bike (Lock)	1. Arrive at locker → open app → "Open Slot 3" 2. App activates camera → scan QR on slot door 3. BLE handshake: app ↔ locker (challenge-response) 4. Solenoid unlocks → vibration + sound feedback 5. User places bike → closes door → weight sensor confirms 6. Locker auto-locks → app shows "Secured until 18:00"
4. Unload Bike (Unlock)	1. Return to locker → app detects geofence (50 m) → shows "Unlock" button 2. Tap "Unlock" → app sends one-time unlock code (expires in 60 s) via BLE 3. Door opens → proximity sensor detects removal 4. App confirms: "Bike released. Session ended."
5. Rent a Bike (from shared pool)	1. Tap "Rent Bike" → see available e-bikes on map 2. Select bike → scan NFC tag on handlebar 3. App unlocks motor lock via BLE 4. Ride starts → app tracks via GPS + accelerometer (low-power mode) 5. End ride: park in any locker → auto-lock → payment deducted
6. Offline Mode	1. No internet → app caches last 10 reservations (encrypted SQLite) 2. BLE still works → user can unlock reserved slot 3. On reconnect: delta sync (only changed records)

LIST OF REQUIREMENTS

Non-Functional Requirements

1. Privacy & GDPR Compliance	
Requirement	How It's Solved
Minimal data collection	Only collect: email (from OAuth), coarse location, reservation timestamps. No precise GPS track unless renting.
User consent	First launch: granular toggles ("Share location for map?" / "Allow analytics?"). Can revoke anytime.
Right to be forgotten	"Delete Account" → app wipes local data + sends DELETE /user to backend → confirmed in < 5 s.
Data encryption	All BLE payloads: AES-256-GCM with per-session keys. Local storage: SQLCipher.
No server-side tracking	Backend logs anonymized hashes only (no IP, no device ID).
2. Low Energy Consumption	
Strategy	Implementation
BLE Low Energy only	Use advertising mode (100 ms interval) when idle. Connection only on user action.
Smart location	Coarse location (Wi-Fi) → geofence trigger (50 m radius) → wake app → switch to fine GPS only if needed.
Sensor duty cycle	Weight & door sensors: sleep 99% of time, wake on BLE command.
Result	< 2% battery drain per 8-hour day
3. Sustainability	
Aspect	Solution
Hardware	Lockers: recycled aluminum, modular battery pack (CR123A, user-replaceable).
E-bikes	Swappable LG 21700 cells, repairable frames.
End-of-life	QR code → "Recycle here" links to local e-waste centers.

LIST OF MATERIALS

Component	Purpose / Function	Qty.	Estimated Unit Price (€)	Est. Subtotal (€)
ESP32 Dev Board	Main IoT controller with Wi-Fi/BLE to send data to Firebase / app	1	10 – 15	10 – 15
Load cell + HX711 *or* **IR distance sensor**	Detects whether a parking slot is occupied	2	7 – 10	14 – 20
Limit switch / magnetic reed sensor	Confirms when the lock is properly closed	2	2	4
RC522 module (or similar)	Reads bike-centric tag for secure access / check-in out	1	8 – 12	8 – 12
Cards / stickers / keyfobs	Linked to the user's bike ID (CICLA registry)	4 – 6	1.5 – 2	6 – 12
Push button switch	Manual check-in/out for prototype demo	1 – 2	1	1 – 2
RGB LED strip (12 V) *or* **high-brightness LEDs**	Indicates slot status: free / occupied / locked	1 strip	10 – 12	10 – 12
Mini servo motor + **simple latch or 3D printed bar**	Demonstrates smart-lock opening / closing	1	8 – 12	8 – 12
Breadboard + jumper wires + resistors kit	Connects and tests components safely	1 set	6 – 8	6 – 8
12 V adapter + **step-down converter to 5 V**	Powers RGB LED strip and ESP32 circuit	1 each	6 – 8	6 – 8
DC jack, Dupont wires, USB cable	General connectivity and power	1 set	3 – 5	3 – 5

→ Estimated total (for 2 bike slots): €76 – 110

ADDITIONAL LINKS

Figjam WorkFile

App Mockup

Github Repository

Github Project Plan